

Spiking Neural Networks as Linear Systems with Control: DMDc Identification, Exact Propagators, and Surrogate-Gradient Training on the Spiking Heidelberg Digits

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Code, logs, and pre-registered protocols: <https://github.com/jimeharrisjr/koopman-based-SNN>

Abstract

Networks of leaky integrate-and-fire (LIF) neurons are usually simulated by small-step numerical integration and treated as irreducibly nonlinear objects. We develop, implement, and stress-test the opposite view: between spikes an LIF network is *exactly* linear, and the one discontinuous event — the spike reset — can be folded into the control input of a discrete linear system $x_{t+1} = Ax_t + Bu_t$. This is precisely the model class addressed by Dynamic Mode Decomposition with control (DMDc), which makes the network’s governing equations recoverable from recordings of its activity. We present the theory of this formulation, quantify its benefits (drift-free simulation at cost parity with a conventional simulator — measured at $1.02\times$ — and operator identification to below 10^{-8} relative error), and give an unusually candid account of its pitfalls: closed-loop identifiability and excitation requirements, the unsuitability of raw spike trains as observables, the bilinearity of hard resets, the limits of spectral gradient arguments, and two pre-registered *negative* results on lifted linear surrogates of genuinely nonlinear neuron models. Finally, we show that the exact formulation supports practical learning: hand-rolled surrogate-gradient backpropagation through time trains recurrent LIF networks on the Spiking Heidelberg Digits (SHD) from 50.2% to 88.6% test accuracy (best single seed) over a five-round experimental campaign, with a variance-reduced ensemble result of 88.2%. A seed audit shows a ± 2.7 -point spread across weight initializations, which retroactively invalidates every sub-3-point margin in the campaign; the surviving effects are recurrence (+12.8 points), augmentation $\times$ training budget (+9.6), and depth (+1.3). All experiments run single-threaded on a laptop CPU and every number reported here has a log file in the repository.

1 Introduction

Spiking neural networks (SNNs) replace the dense synchronous arithmetic of conventional deep learning with sparse, asynchronous events. Their promise — energy budgets orders of magnitude below dense networks, and a model class closer to biological computation — is well documented, but so are their costs: simulation by small-step numerical integration accumulates drift, and training through a discontinuous spike nonlinearity requires approximations (Neftci, Mostafa & Zenke, 2019).

This paper develops one organizing idea and follows it to its limits. The sub-threshold dynamics of the standard LIF neuron are a linear ODE with a closed-form solution; the only genuinely nonlinear event is the threshold crossing. If the spike *reset* is treated as a known impulsive control input rather than as a modification of the dynamics, the entire network becomes a discrete-time linear system with control,

$$x_{t+1} = A x_t + B u_t, \quad s_t = \Theta(v_t - \theta),$$

in which the threshold comparison decides only *which entries of the input are switched on* and never touches the state’s evolution. This is exactly the model class of Dynamic Mode Decomposition with control (DMDc; Proctor, Brunton & Kutz, 2016), the data-driven arm of Koopman operator theory (Koopman, 1931; Mezić, 2005; Brunton, Budišić, Kaiser & Kutz, 2022). Three consequences follow, and each is tested in this paper:

1. **Exact simulation.** The linear part advances by a closed-form matrix exponential; nothing drifts, ever (§2.3). A permanent test in the accompanying Rust library (`kdmd-snn`, built on the `koopman-dmd` crate) requires spike-for-spike agreement with a conventional reference simulator over 1,000 steps, and a benchmark measured the exact engine at 1.02× the reference’s cost — exactness is free, not merely fast.
2. **Identification from data.** Because the network honestly *is* linear plus control, DMDc can recover its operator from recordings (§3). With the input matrix pinned to its known value, the library’s permanent oracle test recovers A to $\leq 10^{-8}$; a fresh numerical replication in this paper reaches machine precision (2.5×10^{-15} relative Frobenius error) and maps out where identification fails.
3. **Trainable exactness.** Surrogate-gradient backpropagation through time, with the surrogate as the *only* approximation in the training loop, trains these networks on real neuromorphic data (§5).

The paper is organized to be as useful to a skeptic as to an enthusiast. Section 2 gives the theory. Section 3 presents identification, including a quantitative study of when it breaks. Section 4 collects every pitfall and caveat we encountered, including two pre-registered experiments that *failed* and are reported as negative results. Section 5 reports the SHD training campaign, including its own audit. Section 6 discusses what the linear-plus- control view does and does not buy relative to the recent spiking state-space-model literature.

A note on method: this project pre-registered its falsifiable claims with frozen thresholds before running the deciding experiments, subjected results to adversarial review, and retained the failures in the record. Where a claim below is enforced by a permanent test in the repository, we say so.

2 Theory

2.1 The LIF neuron and its exact discretization

The current-based LIF neuron carries two state variables: a membrane voltage v and a synaptic current i , with dynamics

$$\tau_m \dot{v} = -(v - v_{\text{rest}}) + R i, \quad \tau_s \dot{i} = -i + \sum_j w_j s_j(t),$$

where incoming spikes s_j deposit charge weighted by w_j , and a spike is emitted when v crosses the threshold θ (we normalize $v_{\text{rest}} = 0$). Both equations are linear. Stacking $x = [v, i]^\top$, the sub-threshold flow over a step of length h has the closed-form solution

$$x_{t+1} = A_{\text{loc}} x_t, \quad A_{\text{loc}} = \begin{bmatrix} \alpha & \gamma \\ 0 & \beta \end{bmatrix}, \quad \alpha = e^{-h/\tau_m}, \quad \beta = e^{-h/\tau_s},$$

with γ the exact current-to-voltage coupling accumulated over the step (the $(1, 2)$ entry of the matrix exponential of the continuous generator). This is not an approximation of the physics; it *is* the solved physics, valid at any step size. The zero in the lower-left corner records the one-way causal structure of the model: current drives voltage, voltage never drives current.

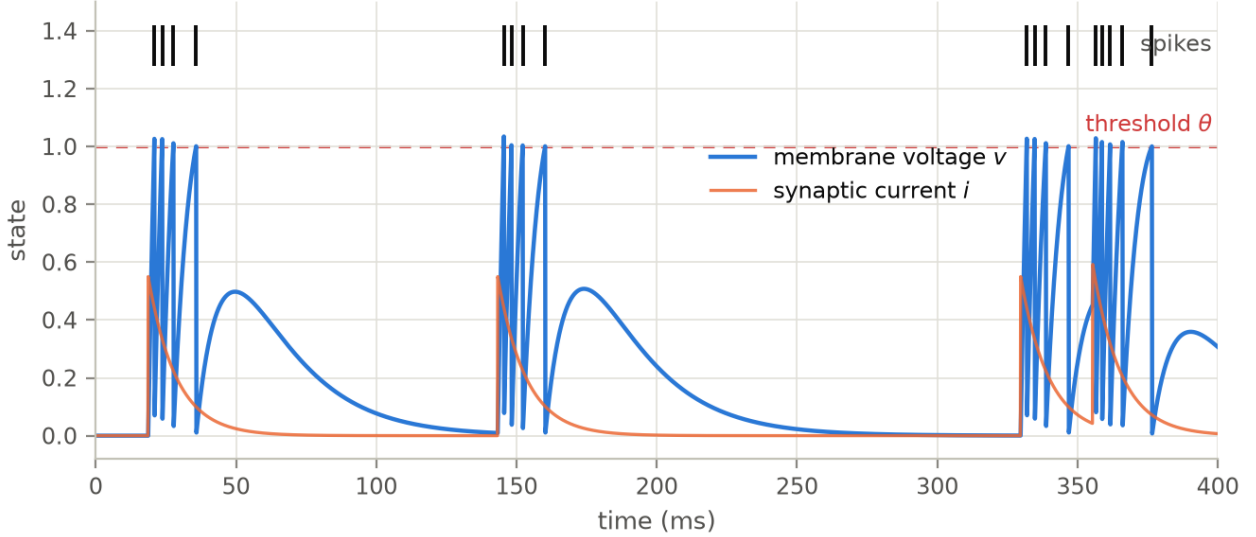


Figure 1. A single LIF neuron simulated with the exact propagator ($\tau_m = 20$ ms, $\tau_s = 10$ ms, $\theta = 1$, Poisson input). The voltage integrates the synaptic current, fires when it touches the threshold, and is reset subtractively; between input bursts both states decay exponentially. Every point of this trace is produced by repeated multiplication with one fixed 2×2 matrix.

2.2 The network, and the reset as a control input

A layer of N identical neurons has state dimension $2N$ and the block-diagonal propagator $A = I_N \otimes A_{\text{loc}}$, so one step costs $O(N)$, not $O(N^2)$. Neurons interact only through spikes, which enter through the control term. Two known linear maps assemble B : the synaptic weight matrix W routes presynaptic spikes into postsynaptic currents, and the *subtractive reset* routes each neuron's own output spike back onto its own voltage with the fixed coefficient $-\theta$:

$$x_{t+1} = A x_t + B u_t, \quad u_t = \begin{bmatrix} s_t^{(l-1)} \\ s_t^{(l)} \\ s_{t-1}^{(l)} \end{bmatrix},$$

where $s_t^{(l-1)}$ are the previous layer's spikes and $s_{t-1}^{(l)}$ the layer's own spikes from the previous step. Subtracting a constant from one entry of the state is a linear operation, so after this move the dynamics are exactly linear-plus-control at every step, spikes included. The only remaining non-linearity is the threshold comparison $s_t = \Theta(v_t - \theta)$ — a decision, not an approximation (Figure 2). Recurrent connectivity adds a second learned matrix W_{rec} routing the layer's own previous-step spikes into its currents through the same door.

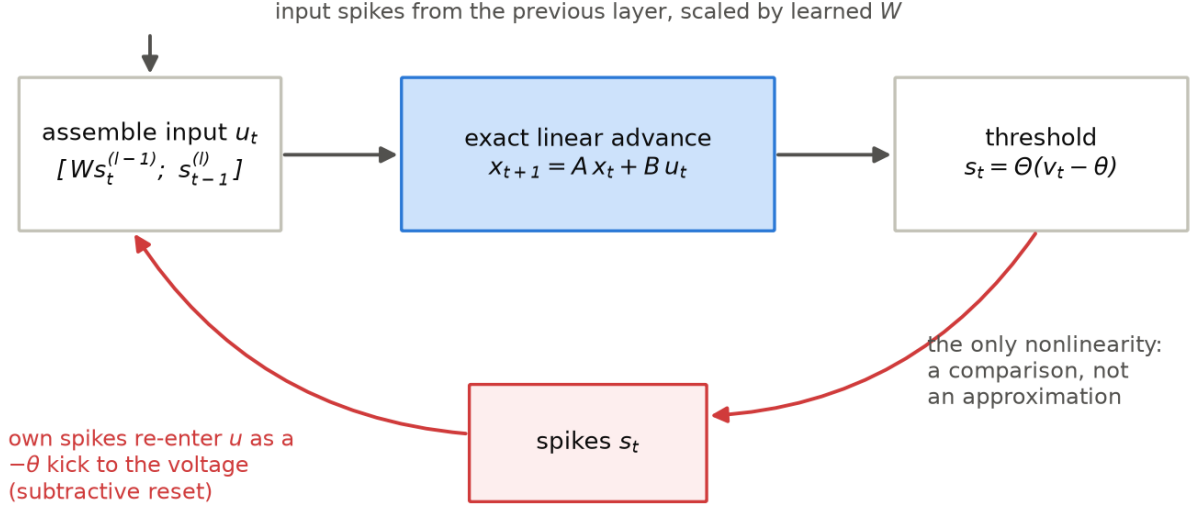


Figure 2. One time step of the formulation. The state advances through the exact linear propagator; a threshold comparison emits spikes; each spike re-enters the input vector at the next step both as drive to downstream neurons (through learned weights) and as a fixed $-\theta$ reset kick to its own voltage.

Two remarks keep this honest. First, the construction requires the *subtractive* (soft) reset. The hard reset ($v \mapsto v_{\text{reset}}$) multiplies the state by the spike indicator and is bilinear in (s_t, x_t) — it cannot be written as $Ax + Bu$ with state-independent matrices (§4.4). Second, the same trick extends to spike-triggered adaptation: an adaptive LIF (AdLIF) neuron adds one more exponentially decaying state variable whose spike-triggered increment is just another fixed jump carried by u , enlarging A_{loc} to 3×3 while preserving exact linearity between spikes. The library implements both under the same spike-for-spike equivalence tests.

In Koopman-theoretic terms (§2.4) the situation is unusually clean: the observables that linearize the flow are the state variables themselves, and the one discontinuity is exiled into the input channel rather than being approximated by a finite-rank operator across the jump — the same design choice as the “linear dynamics + isolated nonlinearity” predictors of Korda & Mezić (2018b).

2.3 What exactness is worth

Figure 3 quantifies the claim against the standard alternative. Panel (a) integrates a sub-threshold neuron with forward Euler at three step sizes and with the closed-form propagator, against the analytic solution: Euler error scales with h as expected, while the propagator sits at the double-precision floor ($\sim 10^{-16}$) at the coarsest step size. Panel (b) shows the consequence for spike timing: under a constant supra-threshold drive (threshold logic identical in all runs, so all differences come from flow integration), Euler at $h = 1$ ms accumulates a phase error of a full inter-spike interval within 49 spikes — the spike train slips by 25 ms over one second — while the propagator, by construction, has zero drift at the same per-step cost.

The library’s benchmark gate (repository, docs/09) completes the cost side: at $N = 1024$ neurons and 100 steps, the structured exact engine measured 3.61 ms per trial against 3.52 ms for a conventional reference simulator — a ratio of 1.02 $\times$, parity within noise — while producing bit-identical

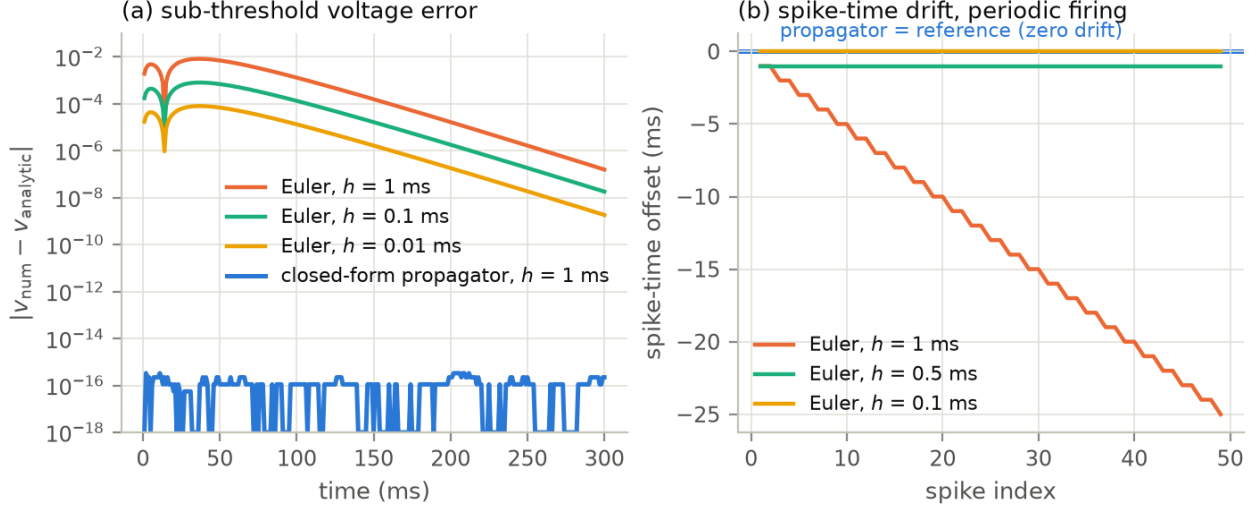


Figure 3. (a) Sub-threshold voltage error against the analytic solution: forward Euler at $h \in \{1, 0.1, 0.01\}$ ms versus the closed-form propagator at $h = 1$ ms, which sits at the floating-point floor. (b) Spike-time offset versus the exact reference under periodic firing: coarse Euler drifts by a full period over one second; the finer steps buy accuracy at 10–100 $\times$ the cost.

spike trains (state agreement $\leq 10^{-12}$ relative over 1,000 steps, enforced by a permanent equivalence test). Exactness costs nothing; what it buys is a simulation that cannot drift and a training Jacobian whose linear part is exact (§5.1). The same benchmark also recorded the mirror-image caveat: a *dense fitted* operator used naively for inference ran at 274 $\times$ the reference cost and was demoted by the project’s pre-registered kill criterion — density is what the block-diagonal structure exists to avoid (§4.1).

2.4 Koopman operators, DMD, and DMDc

For a dynamical system $x_{t+1} = F(x_t)$, the Koopman operator $\mathcal{K}$ acts on observables g by composition, $(\mathcal{K}g)(x) = g(F(x))$, and is linear even when F is not (Koopman, 1931; Mezić, 2005). The price is dimension: a finite-dimensional $\mathcal{K}$ -invariant subspace containing the state observables exists only for special systems (Brunton et al., 2022). Dynamic Mode Decomposition (DMD) is the standard finite-data approximation: given snapshot matrices $X = [x_0 \dots x_{m-1}]$ and $Y = [x_1 \dots x_m]$, it computes the best-fit linear operator $A = YX^\dagger$ via a (truncated) SVD of X (Kutz, Brunton, Brunton & Proctor, 2016), and Extended DMD lifts the state through a dictionary of observables before regressing (Williams, Kevrekidis & Rowley, 2015), converging to a projection of $\mathcal{K}$ as data and dictionary grow (Korda & Mezić, 2018a).

DMDc extends the regression to actuated systems (Proctor et al., 2016): stacking inputs $\Upsilon = [u_0 \dots u_{m-1}]$, it solves

$$[A \ B] = Y \begin{bmatrix} X \\ \Upsilon \end{bmatrix}^\dagger$$

jointly (the Koopman-with-inputs framework of Proctor, Brunton & Kutz, 2018, and the lifted linear predictors of Korda & Mezić, 2018b, generalize this to lifted observables), or — when the input

matrix is known — the better-posed reduced problem

$$A = (Y - B \Upsilon) X^\dagger.$$

The known- B variant is the one this project defaults to, for reasons that §3.2 makes quantitative: in a spiking network the reset column of B is $-\theta$ *by construction* and the feed-forward columns are the known weights W , so fitting them would add estimation variance in exchange for information we already possess — and, in the joint formulation, would expose the regression to the closed-loop collinearity between the state and the network’s own spikes.

For the plain LIF network, then, Koopman theory contributes a *viewpoint* rather than a *lifting*: the linearizing observables were already in hand, and DMDc supplies the identification machinery that this viewpoint licenses. The DMDc solver required for this work did not previously exist in the Rust ecosystem and was contributed upstream to the open-source `koopman-dmd` crate (released in version 0.2.0).

3 Identifying a spiking network from its trajectories

3.1 The oracle result

The library’s identification gate is an oracle test: generate trajectories from a network with known (A, B) — *full spiking trajectories*, resets included, with no masking of spike steps — hand the recording to known- B DMDc, and require the recovered $\hat{A}$ to match the truth within 10^{-8} . This test runs on every build. For this paper we additionally replicated the experiment in an independent numpy implementation ($N = 8$ neurons, 16 state variables, Poisson-probed input, 4,000 snapshots): the noiseless known- B recovery error is 2.5×10^{-15} relative Frobenius — machine precision — confirming that the subtractive-reset-as-control formulation is not merely a modeling convenience but an *identifiable* description of the spiking dynamics.

3.2 When identification fails — measured

The textbook warning for DMDc is that inputs must be exogenous, and a spiking network violates this: its own spikes are deterministic state feedback, $u_t \supset \Theta(Cx_t)$ (Proctor et al., 2016, flag exactly this closed-loop caution). We quantified the practical consequences (Figure 4).

Three findings, in decreasing order of practical importance:

1. **Persistency of excitation dominates everything.** Recording a network under its natural, unvaried drive is not enough: with a constant input the trajectory collapses onto a low-dimensional attractor, the snapshot matrix becomes numerically rank-deficient ($\text{cond}(X) \approx 10^{19}$ versus $\sim 10^3$ for a probed recording), and *both* estimators fail at $O(1)$ error — pinning B does not rescue a recording that contains no information about most of the state space (probed recordings measured $\text{cond}(X) \sim 10^3$ – 10^4). Identification protocols must inject exploratory input (frozen-noise probe currents in our experiments).
2. **The closed-loop threat is real but subtler than the theory suggests.** The spike train is strongly — but not exactly — collinear with the state: regressing the spike indicators on the state explains $R^2 \approx 0.74$ of their variance in our bursty-drive runs, and the stacked regressor matrix $[X; \Upsilon]$ is accordingly ill-conditioned ($\text{cond} \sim 10^{18}$ versus $\sim 10^3$ for X alone). Because the threshold indicator is *binary*, exact rank deficiency is avoided and the joint fit often

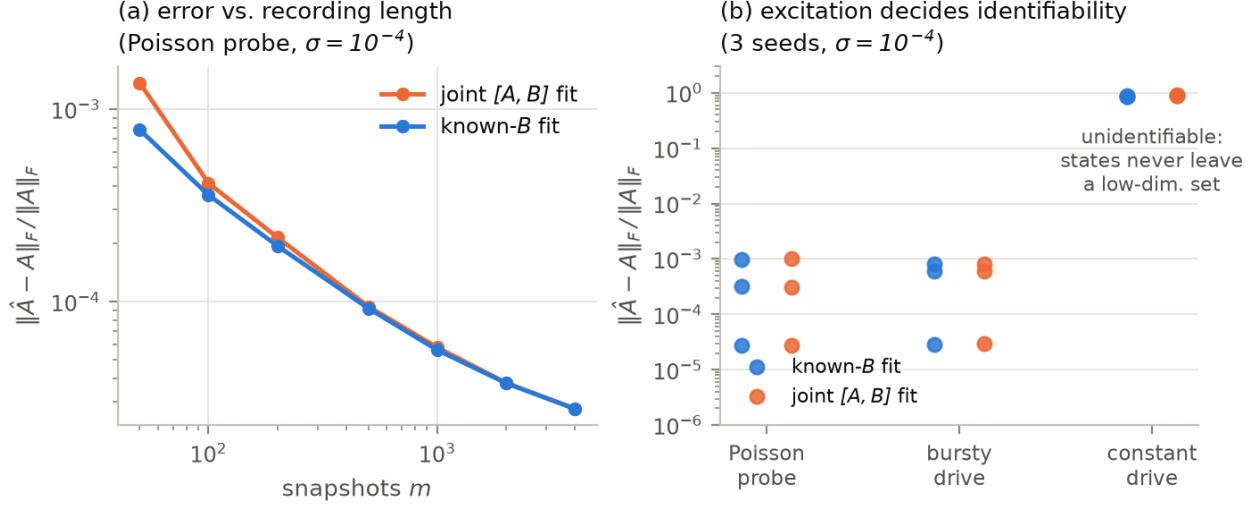


Figure 4. Identification of A from spiking trajectories ($N = 8$ LIF neurons, state dimension 16, measurement noise $\sigma = 10^{-4}$). (a) With a Poisson probe input, both the known- B fit and the naive joint $[A, B]$ fit converge with recording length; known- B is consistently more sample-efficient (1.7 $\times$ lower error at 50 snapshots). (b) The dominant failure mode is *excitation*, not attribution: under a constant drive the state never leaves a low-dimensional set ($\text{cond}(X) \sim 10^{19}$, versus 10^3 – 10^4 when probed) and no method recovers A (relative error ≈ 0.9 for both), while probed recordings identify it to $\sim 10^{-4}$.

survives in practice — but its safe operating window is far narrower. In one bursty-drive configuration, truncating the SVD at 10^{-3} of the top singular value (a routine noise-suppression choice) left the known- B fit at machine precision while the joint fit acquired a 2.2×10^{-2} systematic error *at zero noise* — a bias that no amount of additional data removes, because the truncated directions are exactly where the state/spike attribution is decided. The same truncation can also harm the known- B fit whenever $\text{cond}(X)$ itself exceeds the truncation threshold, which returns us to finding 1.

3. **Known- B is strictly preferable whenever B is known.** It halves the parameter count (1.7 $\times$ lower error at 50 snapshots in Figure 4a), removes the attribution ambiguity by construction, and — since the reset coefficient and the weights are known exactly in this architecture — discards no information. The library exposes both modes and documents the trap.

Beyond recovery of A , the identified operator supports the analyses that motivate the Koopman view: its eigenvalues give per-mode timescales $\tau_j = -h / \ln |\mu_j|$, stability margins $1 - |\mu_j|$, and oscillation frequencies, and the DMDC output basis yields rank- r reduced-order stepping of a recurrent layer at $O(Nr)$ cost — validated in the library within 10% rollout RMSE in the sub-threshold regime (the spiking-regime extension is open; §6).

4 Pitfalls and caveats

This section collects everything we know that a reader attempting this method should be warned about. Items 4.1–4.5 are analytical; 4.6 reports two pre-registered experimental failures; 4.7–4.8 concern training practice.

4.1 DMD does not “linearize” the LIF network — the linearity was already there

The most tempting misstatement of this work is that Koopman/DMD machinery *makes* the network linear. For current-based LIF with subtractive reset, the sub-threshold propagator is known in closed form, per neuron, with no data, no SVD, and no regression; DMD applied to such a network can only re-estimate known decay constants with sampling error. A substantial 2023–2026 literature (PSN, Fang et al., 2023; SpikingSSMs; P-SpikeSSM; SPiKE-SSM; SiLIF) already builds SNN training directly on the linear-state-space form of LIF without Koopman language, and constitutes the baseline any Koopman-flavored claim must beat. The genuine contributions of the operator view are elsewhere: identification (§3), reduced-order modeling, spectral diagnostics, and — in principle — lifted surrogates of genuinely nonlinear neurons, for which see the negative results in §4.6. Our benchmark’s demotion of the dense fitted-operator inference path ($274\times$ the structured engine’s cost, §2.3) is the same point made economically: a dense identified A can never compete with a structure-exploiting update for homogeneous layers.

4.2 Closed-loop identification and excitation

Treated quantitatively in §3.2. Summarized as advice: never fit B when you can pin it; never trust a fit on a recording whose input did not explore; and check $\text{cond}(X)$ before believing $\hat{A}$, because rank truncation chosen to suppress noise will silently truncate the *dynamics* when the data are unexciting.

4.3 Binary spike trains are bad observables

DMD regression assumes observables that evolve approximately linearly. Raw spike indicators do not: they are discontinuous in the state and carry no magnitude information. Every successful application of DMD to neural data uses continuous signals (e.g., ECoG voltages with delay embedding; Brunton, Johnson, Ojemann & Kutz, 2016). In this project all identification operates on membrane voltages and synaptic currents; spikes appear only in the input channel Υ , where their discreteness is harmless because B is linear in them by construction.

4.4 Hard reset is not linear in the control

The subtractive reset ($v \leftarrow \theta$) is a state-independent jump and folds into Bu . The hard reset ($v \mapsto v_{\text{reset}}$) is $v_{t+1} = (1 - s_t)(\alpha v_t + \dots) + s_t v_{\text{reset}}$ — bilinear in (s_t, v_t) , hence outside the $Ax + Bu$ class with constant matrices. The Spike Response Model (Gerstner & Kistler, 2002) reaches the same conclusion from the convolutional side: its reset kernel formulation is precisely the soft reset. Libraries and analyses in this framework should default to the subtractive reset; hard-reset models require either a bilinear correction term or acceptance of approximation error at every spike.

4.5 The spectrum diagnoses gradients; it does not fix them

An early version of this project’s premise claimed that the exact linear propagator “prevents exploding or vanishing gradients.” The literature does not support *prevents*, and neither do we. Gradient norms through the linear part scale as $\rho(A)^T$ (Pascanu, Mikolov & Bengio, 2013), and any leaky — that is, dissipative — network has $\rho(A) < 1$: that is what leak *means*. Gradients through the linear part of an LIF network vanish geometrically over long horizons exactly as they do in any conventional simulation, and the surrogate-derivative factors at threshold crossings (§5.1) are typically

the dominant source of gradient attenuation in deep SNNs (Neftci et al., 2019). What the operator view actually offers is *diagnosis and design*: the eigenvalues are directly readable, per-mode credit-assignment horizons $\tau_j = -h / \ln |\mu_j|$ can be reported, and the spectrum can in principle be constrained toward the unit circle as an explicit regularizer, in the spirit of unitary-RNN methods. That is a real capability; it is not automatic prevention, and we flag the distinction because we initially got it wrong.

4.6 Lifted linear surrogates of nonlinear neurons: two pre-registered failures

For genuinely nonlinear neuron models — Izhikevich’s quadratic model (Izhikevich, 2003), AdEx (Brette & Gerstner, 2005) — no closed-form linear propagator exists, and EDMD lifting is the principled hope, with an established record on smooth neuron-type dynamics (e.g., the isostable analysis of excitable FitzHugh–Nagumo dynamics; Mauroy, Mezić & Moehlis, 2013). We pre-registered two experiments with frozen accuracy gates before running them (repository, docs/04 and docs/07). Both failed their gates, and the failure modes are more instructive than a success would have been (Figure 5).

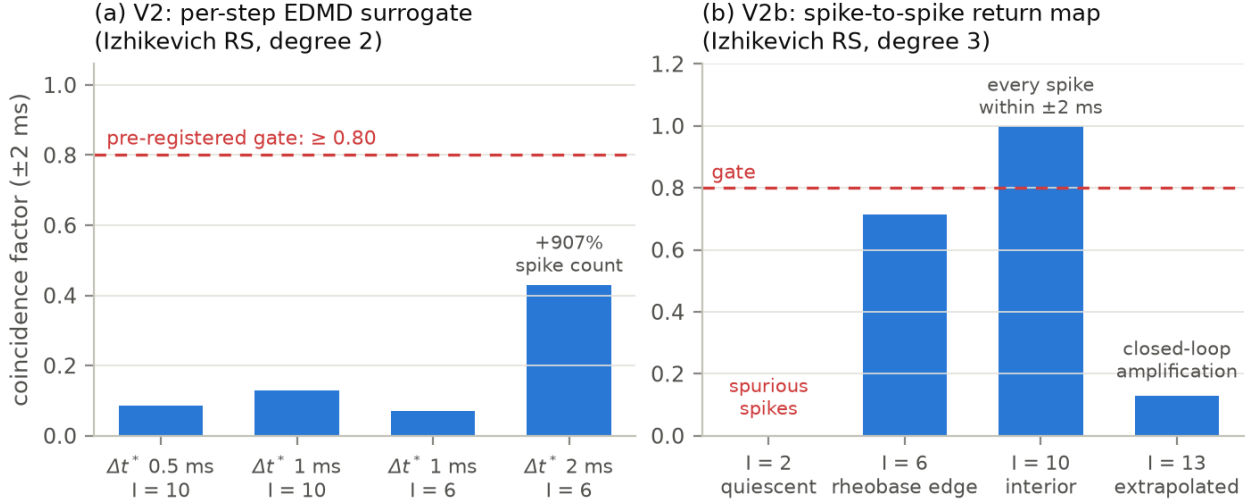


Figure 5. Two pre-registered negative results on lifted surrogates of the Izhikevich regular-spiking (RS) neuron, replotted from the recorded gate tables (docs/05, docs/08). (a) The per-step EDMD surrogate (V2) nearly matches spike *counts* and first-spike latency but fails the ± 2 ms coincidence gate at every registered configuration: a 4–9% per-interval period error compounds into unbounded phase drift (the $\Delta t^* = 2$ ms column “improves” coincidence only by emitting 9× too many spikes). (b) The spike-to-spike return-map surrogate (V2b) is *perfect* at the interior training current ($I = 10$: every spike within ± 2 ms over a full second) and fails at the envelope’s edges — spurious spikes below rheobase, a near miss at the rheobase edge, and closed-loop error amplification under extrapolation.

V2 — per-step EDMD surrogate. A polynomial-dictionary EDMD fit of the Izhikevich flow passed its sub-rheobase (quiescent) gates and failed every spiking gate: coincidence factors of 0.07–0.13 against a pre-registered gate of 0.80, with spike counts and first-spike latency nearly correct. The signature is cumulative phase drift — the same disease Euler integration exhibits in Figure 3b, now caused by dictionary misfit on the spike upstroke/reset re-entry region rather than by step size. Degree-3 dictionaries achieved *better* one-step residuals and then diverged at rollout ($\rho(K)$ up

to 350) — the classic EDMD overfit-to-instability trade. Notably, the experiment’s cost predictions held: coarse Euler failed the same ± 2 ms window (P2 confirmed), so the surrogate did not lose on cost — it lost on its own accuracy.

V2b — spike-to-spike return map. The bounded rescue attempt regressed the inter-spike interval map directly, making per-cycle error the fitted quantity. Inside the training envelope it worked *perfectly* — coincidence 1.000 for both gated neuron types at $I = 10$, something V2 never approached — and failed at the edges through three specific, pre-registered mechanisms: the approach-flow (time-to-first-spike) map is much harder than the on-cycle map (23–58% held-out error versus a 5% precondition); quiescence prediction fails as a censored-data problem (a map trained only on spiking segments happily predicts finite latencies below rheobase); and teacher-forced accuracy did not transfer to closed-loop rollout at the extrapolation current (-0.31% interval bias amplifying to a 0.13 coincidence), a structural obstacle we recorded as closed-loop amplification. The registered consequence was executed: the nonlinear-surrogate track closed, and the library’s fitted-operator value cases reduced to reduced-order modeling and spectral diagnostics.

The general lesson for Koopman-style modeling of spiking systems: finite-rank linear surrogates across a spike discontinuity behave like regressions through a jump — they average over it. Event-level reformulations (fitting the map *between* spikes) can be locally exact but inherit an extrapolation cliff at the edges of their training envelope. Anyone proposing a lifted spiking surrogate should be required to state its validity envelope and its behavior under closed-loop rollout, because interior interpolation accuracy predicts neither.

4.7 Fitted operators go stale under training

A DMD-fit $\hat{A}$ of a *recurrent* layer absorbs W_{rec} ; the moment training updates the weights, the fit describes the previous network. Any scheme that trains through a fitted operator needs a re-fitting schedule (alternation, or streaming DMD), and the literature contains no ready-made answer for the SNN case. This project sidesteps the problem by training only through the *closed-form* operator, which depends on no data and can never go stale — but that option exists only because plain LIF is exactly linear, which is also why the fitted operator adds no inference value there (§4.1). The tension is structural: where the fitted operator is trivial it is also correct, and where it would be valuable it goes stale.

4.8 Seed noise, evaluation discipline, and pre-registration

Section 5.4 measures a ± 2.7 -point test-accuracy spread across three weight initializations of an *identical* training recipe on SHD. This single number retroactively reframes the campaign: every margin below roughly three points, anywhere, is individually inconclusive. We consider the following practices non-optional for SNN benchmark work, all of them cheap: fix the data order and seeds so that any two runs differ only in the variable under test; evaluate on the full test set, never a sub-sample; re-run the best recipe under new seeds before believing it; ensemble across seeds when reporting a headline number, and report the spread; and log the losers. Every accuracy figure in Section 5 has a corresponding log file in the repository, including the failures.

5 Training on the Spiking Heidelberg Digits

5.1 Learning machinery

The threshold Θ has no usable derivative, so training uses the surrogate-gradient method (Neftci et al., 2019): the forward pass keeps the hard step — the spikes are real spikes — while the backward pass substitutes a smooth pseudo-derivative at the threshold (Figure 6). We use the SuperSpike fast sigmoid $\sigma'(x) = (1 + \beta|x|)^{-2}$ (Zenke & Ganguli, 2018) evaluated at the pre-reset potential; exponential-kernel (Shrestha & Orchard, 2018), arctan, and boxcar surrogates are implemented as alternatives. Backpropagation through time unrolls the network over all input bins; because the sub-threshold step is the exact propagator of §2.1, the Jacobian of the linear part is exact, and the surrogate is the *only* approximation in the training loop. The gradients are implemented by hand in Rust with no autograd framework and are validated against finite differences on the smooth path; the readout is a linear map on mean output spike counts, logits = $R \cdot \frac{1}{T} \sum_t s_t$, trained jointly under softmax cross-entropy. Alternatives to unrolled surrogate BPTT exist and would be natural comparison points: e-prop factorizes the gradient into forward-computable eligibility traces (Bellec et al., 2020), and EventProp computes exact gradients of spike-time losses by adjoint methods with jump conditions at spike events (Wunderlich & Pehle, 2021) — the latter intellectually close to this project, since it too treats the between-spike dynamics as exactly integrable.

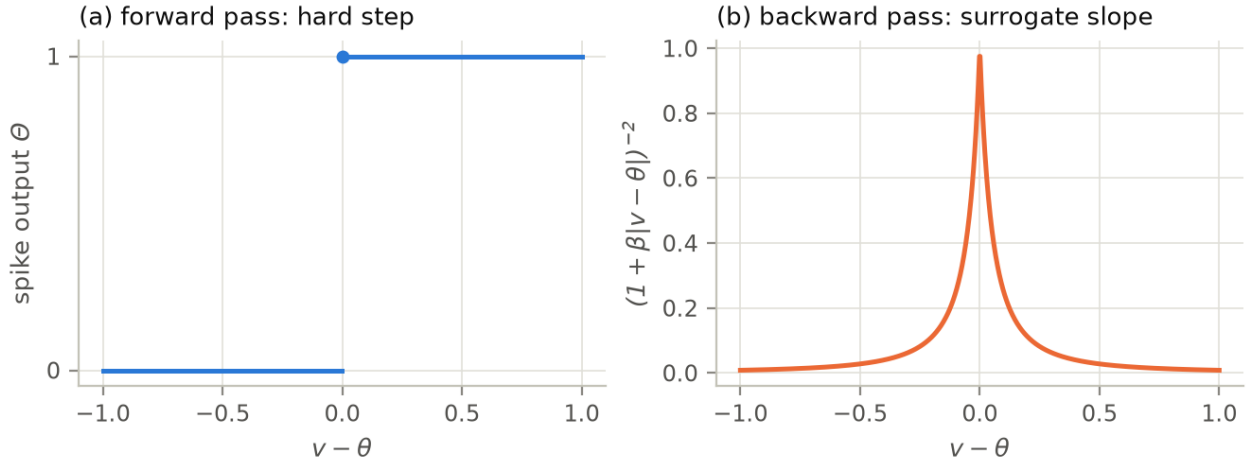


Figure 6. The surrogate gradient. (a) The forward pass keeps the true hard threshold. (b) The backward pass substitutes a smooth slope peaked at the threshold ($\beta = 10$ shown), so weights that nearly caused or nearly prevented a spike receive credit.

5.2 Task and protocol

The Spiking Heidelberg Digits (Cramer, Stradmann, Schemmel & Zenke, 2020) are spoken digits (zero through nine, English and German — 20 classes, chance 5%) encoded into spike trains on 700 channels by a detailed cochlea model: 8,156 training and 2,264 test recordings. Our pipeline pools the channels (2:1 to 350 in the final recipes), bins spikes at 10 ms into 100 steps (1 s), and trains with Adam at 5×10^{-3} , batch 32. LIF parameters are biologically ordinary ($\tau_m = 20$ ms, $\tau_s = 10$ ms). Evaluation uses the largest batch-divisible subset of the test set (2,240 of 2,264 samples) for every reported number. Every run consumes an identical minibatch sequence, with data shuffling and weight initialization seeded separately, so any difference between two runs is attributable to the change under test. The training-set augmentation used from round 3 onward corrupts each

presentation independently: 15% random event dropout, spectral shift of up to ± 25 channels, and $\pm 10\%$ time stretch.

5.3 The campaign

The experimental campaign ran in five pre-planned rounds, each testing a batch of hypotheses against the previous round’s best recipe. Table 1 and Figure 8 summarize the trajectory; Figure 7 shows the two decisive loss-curve comparisons.

Table 1. Campaign milestones (full per-experiment tables: repository, demo/RESULTS.md).

Stage	Configuration change	Test acc.	Cumulative gain
First demo	100 ch, 1×128, 600 minibatches	0.502*	—
Round 1 (I)	350 ch, 1×256, 3,000 minibatches	0.680	+17.8
Round 2 (L)	+ recurrent hidden layer (W_{rec} zero-init)	0.808	+30.6
Round 3 (R)	+ augmentation × 6,000 minibatches	0.873	+37.1
Round 4 (X)	+ second recurrent layer (256–256)	0.886	+38.4
Round 5 (AF)	ensemble ×3 of the round-4 recipe	0.882 (variance-reduced)	+38.0

*The first demo evaluated on a 512-sample subset; all subsequent numbers use the full protocol of §5.2.

Round 1 established an interaction that recurs throughout: input resolution and training budget must move together. Extra budget on coarse 7:1-pooled input *reduced* test accuracy (memorization), while the same budget on 2:1-pooled input paid honestly. Round 2’s single change — recurrent connections grown from an exactly-zero initialization by the through-time gradient — was the campaign’s largest effect at +12.8 points, consistent with the published literature in which recurrence is the known difference-maker on this dataset. Round 3 isolated a second interaction: at 6,000 minibatches the unaugmented recurrent network overfit (train loss 0.038, test 0.777) while the augmented one reached 0.873 — augmentation converts budget from memorization fuel into generalization, and neither ingredient worked alone. Round 4 found that depth pays only once the recipe supports it: the same second layer that had been harmful in round 1’s feedforward setting gained +1.3 points under recurrence and augmentation, while adaptive (ALIF) neurons — dominant on published leaderboards — *lost* accuracy in this configuration (homogeneous: −3.7 points), because the published ALIF wins depend on learned time constants and finer time resolution that this library does not yet implement.

5.4 The audit: seed noise, and what survives it

Round 5 was planned as a push past 0.90 and became an audit instead. Re-running the round-3 recipe *unchanged* except for the weight-initialization seed gave 0.873, 0.819, and 0.858 (mean 0.850,

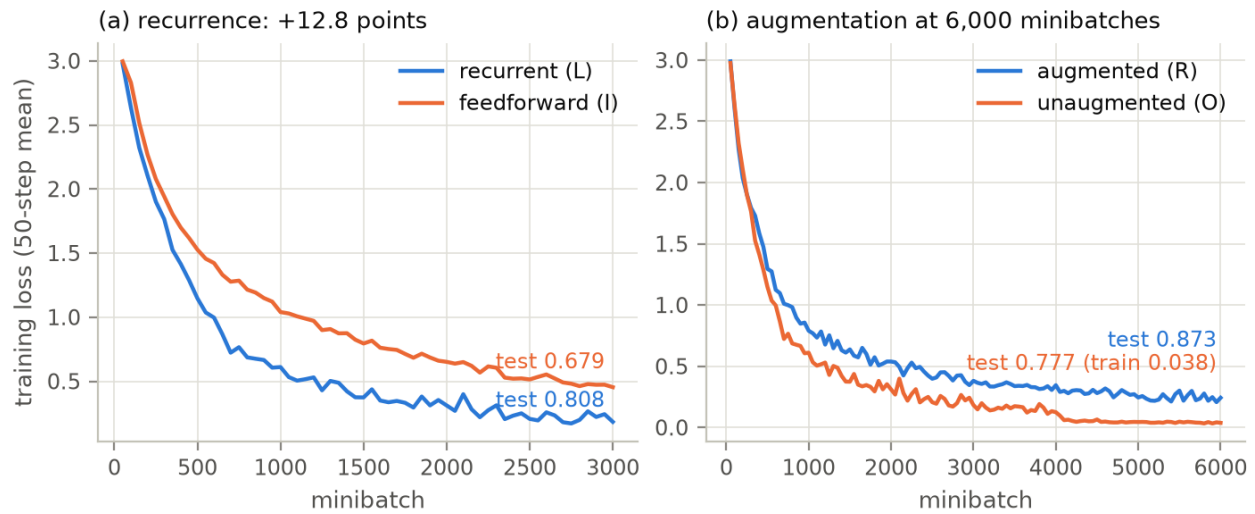


Figure 7. The two decisive comparisons, from the recorded training logs. (a) At equal budget and architecture, the recurrent network (L) separates from its feedforward control (I) within 650 minibatches and finishes 12.8 points higher on test. (b) At 6,000 minibatches, the unaugmented network (O) drives training loss to 0.038 and *loses* 9.6 test points to its augmented twin (R): the loss curves cross in the opposite direction of the test results.

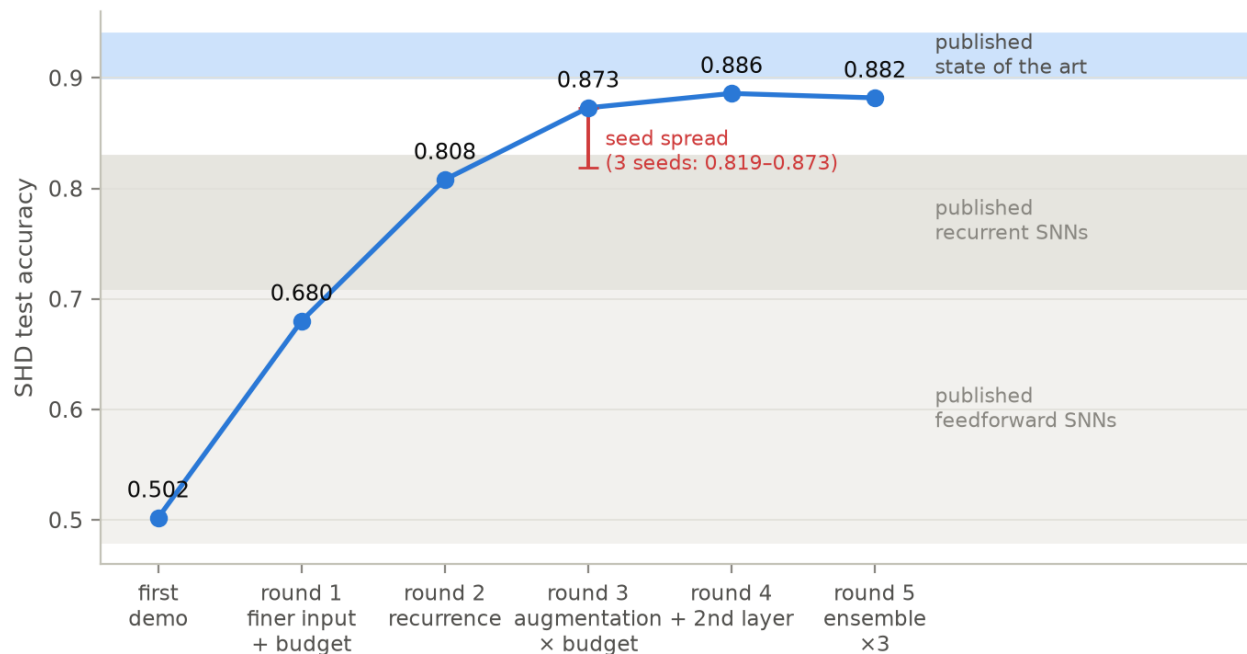


Figure 8. The campaign against the published landscape. Shaded bands mark the approximate published ranges on SHD for feedforward SNNs (0.48–0.71), recurrent SNNs (0.71–0.83), and the augmented/adaptive state of the art (0.90–0.94). The error bar marks the measured three-seed spread of the round-3 recipe (§5.4). Points are best single runs except round 5, which is the three-member ensemble.

spread ± 2.7 points; Figure 9a). The round-3 headline was therefore a lucky draw, and every sub-3-point margin in the campaign — including round 4’s +1.3 for depth as a single comparison, and round 3’s +0.4 for 512-wide-with-decay — is individually inside noise. The axis-level conclusions survive where effects were large (recurrence +12.8, augmentation $\times$ budget +9.6) or replicated across configurations.

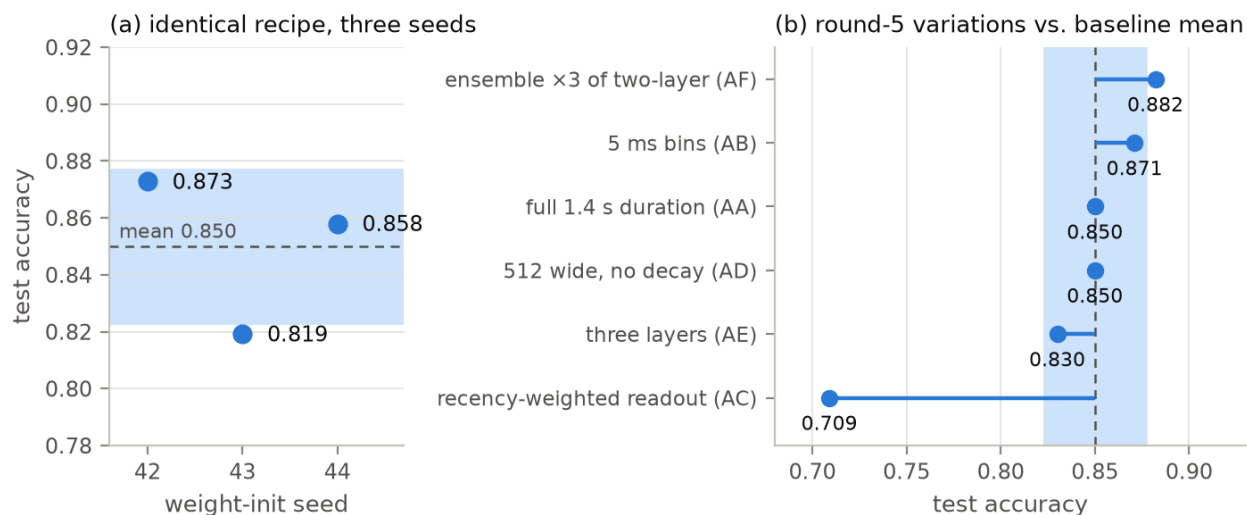


Figure 9. (a) The seed audit: one identical recipe, three weight initializations, spanning 5.4 points. The shaded band (mean ± 2.7) is the noise scale against which all single-run margins must be judged. (b) Round-5 variations against that band: only the ensemble (above) and the recency-weighted readout (far below) are decisively outside it.

The remaining round-5 experiments (Figure 9b) delivered mostly negative but informative results. A third recurrent layer lost ~ 2 points at $1.5\times$ the cost — each depth increment appears to need an enabling ingredient this uniform recipe lacks. Recency-weighting the readout (exponential trace, $\kappa = 0.95$) collapsed accuracy by 14 points, demonstrating that the count readout is load-bearing: evidence for a word is spread across the whole utterance, and any temporal readout must *add* memory rather than replace the integral. Full-duration input and 5 ms bins both landed within seed noise of the defaults. The honest final number is the three-member ensemble of the two-layer recipe at **0.882** — it matches the best single draw (0.886) while averaging away the seed lottery, and its members’ errors are too correlated for voting to buy more than that.

5.5 Where this lands

A network of the plainest possible neurons — subtractive-reset LIF, fixed biologically-generic time constants, count readout — trained by hand-rolled surrogate BPTT on a single laptop thread reaches the top of the published recurrent-SNN band on SHD (~ 0.71 – 0.83) at $0.88 \pm$ noise, with ~ 2.5 hours of training for the final ensemble. The ~ 3.5 -point gap to the published state of the art (0.90 – 0.94) is not mysterious: those systems use learned per-neuron time constants, attention or state-space readouts, and heavier augmentation. Each is a specific, implementable feature — the exact formulation makes learned time constants unusually clean, since the propagator entries are closed-form functions of τ through which one can backpropagate — and none was within this campaign’s fixed scope.

6 Discussion

What the linear-with-control view actually buys. Four things, in the order we can defend them: (i) drift-free simulation at parity cost, with exactness enforceable by test rather than asserted (§2.3); (ii) unbiased identification of network dynamics from recordings, with the reset folded into a known control channel (§3); (iii) spectral transparency — timescales, stability margins, and mode structure of a trained network read directly off the identified operator; and (iv) an exact training Jacobian for the linear part, leaving the surrogate as the only approximation (§5.1). Against the spiking state-space-model literature (Fang et al., 2023, and successors), which occupies the neighboring “linear state + threshold” design point, contributions (ii) and (iii) are the differentiating ones: those works use the linear form for parallel training, not for identification or operator-theoretic analysis.

What it does not buy. It does not linearize anything that was not already linear (§4.1); it does not by itself extend to hard resets (§4.4), conductance-based synapses (bilinear in the state), or genuinely nonlinear neuron models — our two pre-registered attempts at the latter failed in instructive ways (§4.6); it does not repair vanishing gradients (§4.5); and a fitted operator for a layer under training goes stale by construction (§4.7).

Limitations and open problems. The reduced-order value case is currently validated only in the sub-threshold regime; extending rank- r surrogate rollouts to fully spiking activity is a well-posed open problem, as is a principled re-fitting schedule for operators under training. On the learning side, the seed audit (§5.4) bounds what single runs on SHD can claim, and our accuracy ceiling analysis points to learned time constants and a trained temporal readout as the highest-value missing features. On the identification side, our excitation experiments used white-noise probes; the minimal probe richness required for a given network scale is unquantified. Finally, the campaign’s protocol-before-results discipline currently rests on the documents’ internal dating; committing pre-registrations and results in separately hashed commits would make it externally verifiable.

7 Conclusion

Treating a spiking network as what it mathematically is — an exactly linear system punctuated by threshold decisions, with resets riding the control channel — yields a simulator that cannot drift at no extra cost, an identification pathway with oracle-grade accuracy and quantified failure modes, and a training loop whose only approximation is the surrogate gradient. The same investigation, pursued with pre-registered gates, delimits the idea sharply: the operator view adds nothing to what was already linear, fails (so far) to cross the spike discontinuity for genuinely nonlinear neurons, and demands excitation and honesty about seed noise from anyone who uses it. Both halves — the capabilities and the limits — are enforced by tests and logs in the accompanying repository, which we offer as much as a methodological artifact as a software one.

Reproducibility statement

The library, all experiment harnesses, raw logs, pre-registration documents, and this paper’s figure-generation scripts are in the repository. The test suite (`cargo test --release`) enforces the exactness, equivalence, identification-oracle, and gradient claims on every build. The SHD demo and sweep (`cargo run --release --features datasets --example shd_sweep <TAGS>`) reproduce every number in Section 5; raw logs for every reported run are in `demo/`, indexed by `demo/RESULTS.md`. Figures 1, 3, and 4 are generated by `paper/scripts/make_figures.py` (numpy reimplementations of the library’s mathematics, seeds fixed); Figures 7–9 are plotted directly

from the recorded sweep logs; Figure 5 replots the recorded gate tables of docs/05 and docs/08. Runtimes are minutes for the round-1 experiments and up to ~2.5 hours for the final ensemble, single-threaded on an Apple Silicon laptop.

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